ChatGPT

Minimising total treatment time

We have to minimise $T = x_1 + x_2$ subject to the linear constraint $x_1 + a_2\, x_2 = 1$ (with $x_1, x_2 \geq 0$). By the fundamental theorem of linear programming, a linear objective on a linear constraint attains its optimum at a corner (extreme point) of the feasible set [1]. The only corners are $(x_1, x_2) = (1, 0)$ and $(0, 1/a_2)$. At $(1, 0)$ the total time is $T = 1$, while at $(0, 1/a_2)$ it is $T = 1/a_2$. Thus if $a_2 < 1$ then $1/a_2 > 1$ so the minimum is at $(1, 0)$; if $a_2 > 1$ the minimum is at $(0, 1/a_2)$.

The given code makes a_2 extremely close to 1. In fact, evaluating $a_2 = 2f(n) + \frac{1}{2}$ (with the huge exponents shown) shows $a_2 < 1$ (numerically one finds $a_2 \approx 0.9999\ldots$). Hence the optimum is at $(1, 0)$. The infinity-norm distance from $(1, 0)$ to the candidate point $(0.1, 0.9)$ is $\max(|1 - 0.1|, |0 - 0.9|) = 0.9$, which is far larger than 0.15. Therefore $\{0.1, 0.9\}$ is *not* within 0.15 of any minimiser.

Answer: No. The true minimiser is essentially $(1, 0)$, and $(0.1, 0.9)$ lies at distance $0.9 > 0.15$ in the infinity norm. [1]

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[1] Fundamental theorem of linear programming - Wikipedia
https://en.wikipedia.org/wiki/Fundamental_theorem_of_linear_programming